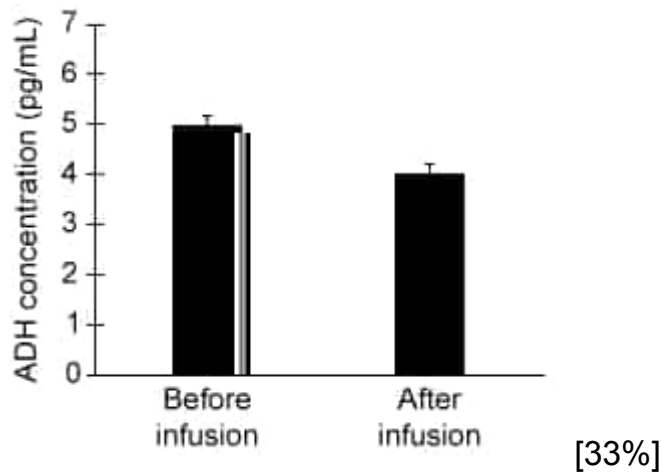
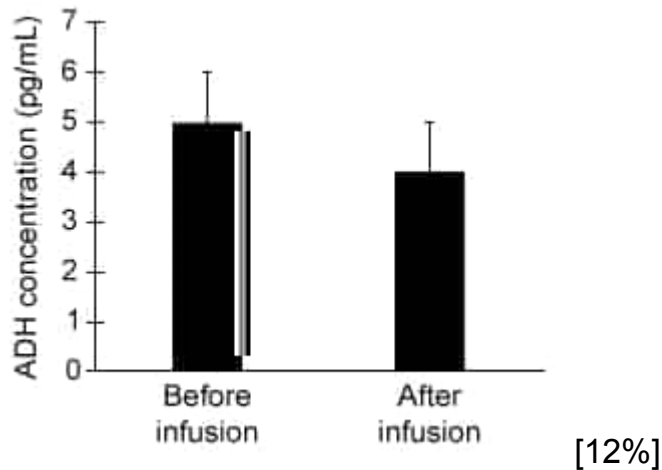


In an experiment, healthy men received an infusion of isotonic saline solution designed to induce a large increase in the blood volume without altering blood osmolarity. Which figure shows the expected effect of the infusion on the antidiuretic hormone (ADH) concentration of the blood?

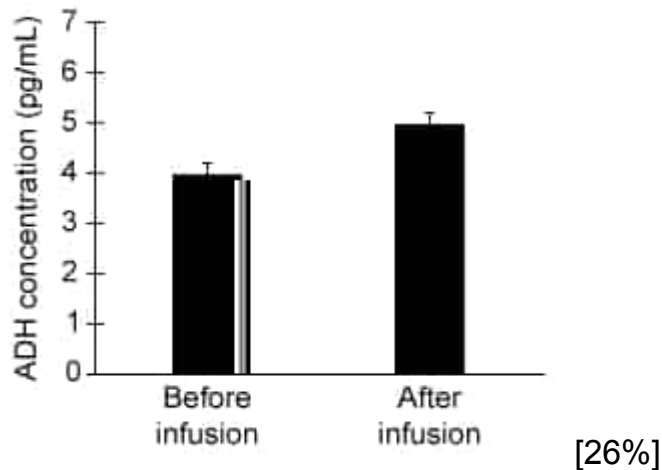
✓ ☒ A.



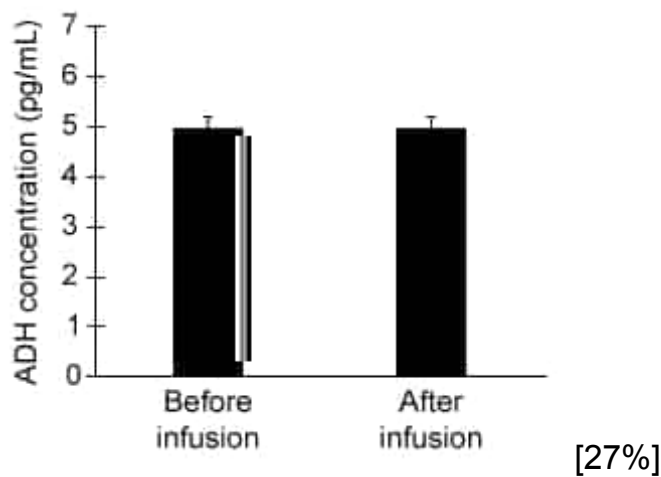
☐ B.



☐ C.



☐ D.



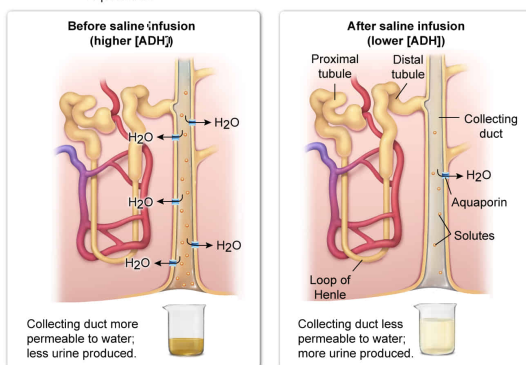
Correct

33% answered correctly

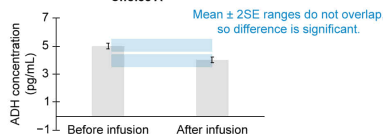
Explanation:

Increased blood volume would reduce antidiuretic hormone (ADH) concentration

Experiment: Measurement of [ADH] before and after saline infusion
Expectation:



Choice A



A large proportion of the body mass is water, which is continually depleted by urine and fecal losses and evaporation from the skin. A small fraction of this water loss is balanced by water released from metabolic reactions (eg, [reduction of oxygen](#)); the remainder of the required intake must be balanced via consumption or infusion.

The body regulates its water content by sensing blood volume, [osmolarity](#), and pressure. Special sensors (ie, stretch receptors) monitor the degree of atrial stretching in the [heart](#) (ie, blood volume) and transmit this information to the [hypothalamus](#). The hypothalamus then signals the [posterior pituitary](#) to adjust antidiuretic hormone (ADH) (vasopressin) secretion accordingly. When blood volume is low, ADH causes the insertion of [aquaporins](#) (water channels) in the [collecting duct](#), which increases water permeability and reabsorption, and ultimately increases blood volume and blood pressure.

The question describes an experiment with a group of men in which their blood volume was increased via [an isotonic](#) saline infusion, and the question requires choosing the graph depicting changes in expected blood ADH concentration. ADH concentration in the blood

is inversely related to blood volume, all other factors being equal. This relationship is consistent with the role of ADH to stimulate water retention and means that ADH release should be suppressed during blood volume expansion to allow more water excretion in urine. Therefore, **blood ADH concentration** would be expected to be **significantly lower after the infusion**.

When researchers compare experimental values from different groups or conditions, they often express the results in terms of **mean or median** values along with the variation in the experimental values (eg, standard deviation, standard error). As a general indicator of whether values from different groups or conditions are significantly different, the ranges defined by the means plus or minus *two* standard errors (which approximates the **95% confidence interval**) are sometimes compared. If these ranges **overlap**, the difference between the values from the different groups or conditions are considered nonsignificant whereas nonoverlapping ranges are considered indicative of a statistical difference, as shown on the graph in Choice A.

(Choices B, C, and D) Blood ADH concentration would be expected to *significantly* decrease (*not* increase or stay the same) after a large increase in blood volume.

Educational objective

Antidiuretic hormone (ADH) is secreted by the posterior pituitary gland in response to reductions in blood volume or osmolarity. This secretion causes increased water reabsorption in the collecting duct.

Time spent:0 sec

Subject:
Biology

QID:404149

System:
Endocrine and Nervous Systems